Project Report - AI-Based Self-Adaptive Construction Scheduling

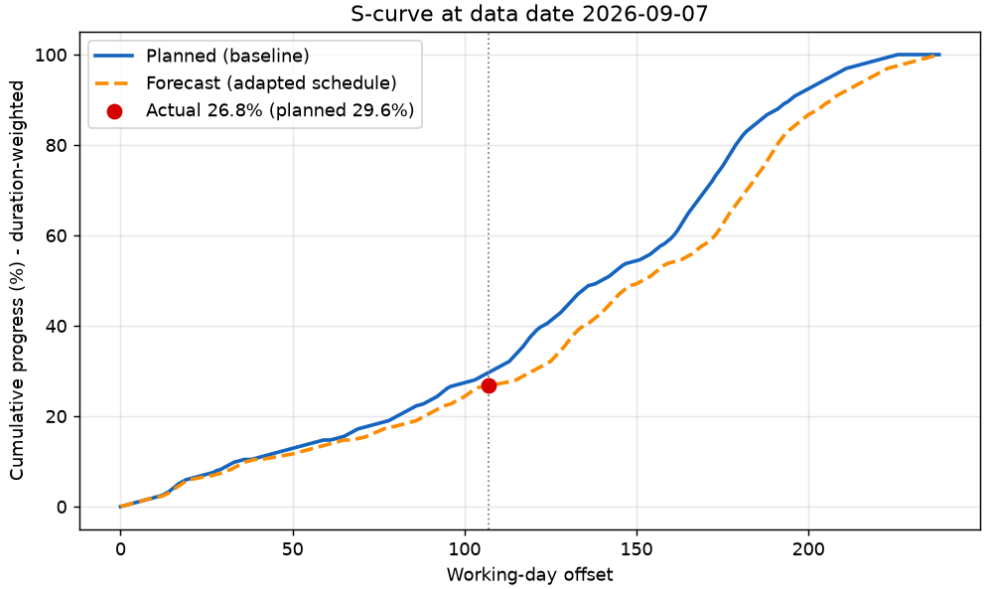
All case-study data and all model training data are SYNTHETIC. Model scores measured on synthetic projects are NOT real-world validation. RII weights are PROVISIONAL (the questionnaire survey has not been conducted).

1. Project summary

Item	Value
Project	G+3 Residential Building (synthetic case study) (CS-G3-001)
Data source	SYNTHETIC
Activities / dependencies	67 / 94
Project start	2026-05-04
Data date	2026-09-07
Baseline completion	2027-02-02 (226 working days)
Current forecast completion	2027-02-16 (238 working days)
Schedule variance	12 working days / 14 calendar days
Adaptation mode / threshold	expected / P(delay) >= 0.60
Composite risk index (provisional RII)	0.687
ML model	LogisticRegression + XGBoost (trained on SYNTHETIC data)

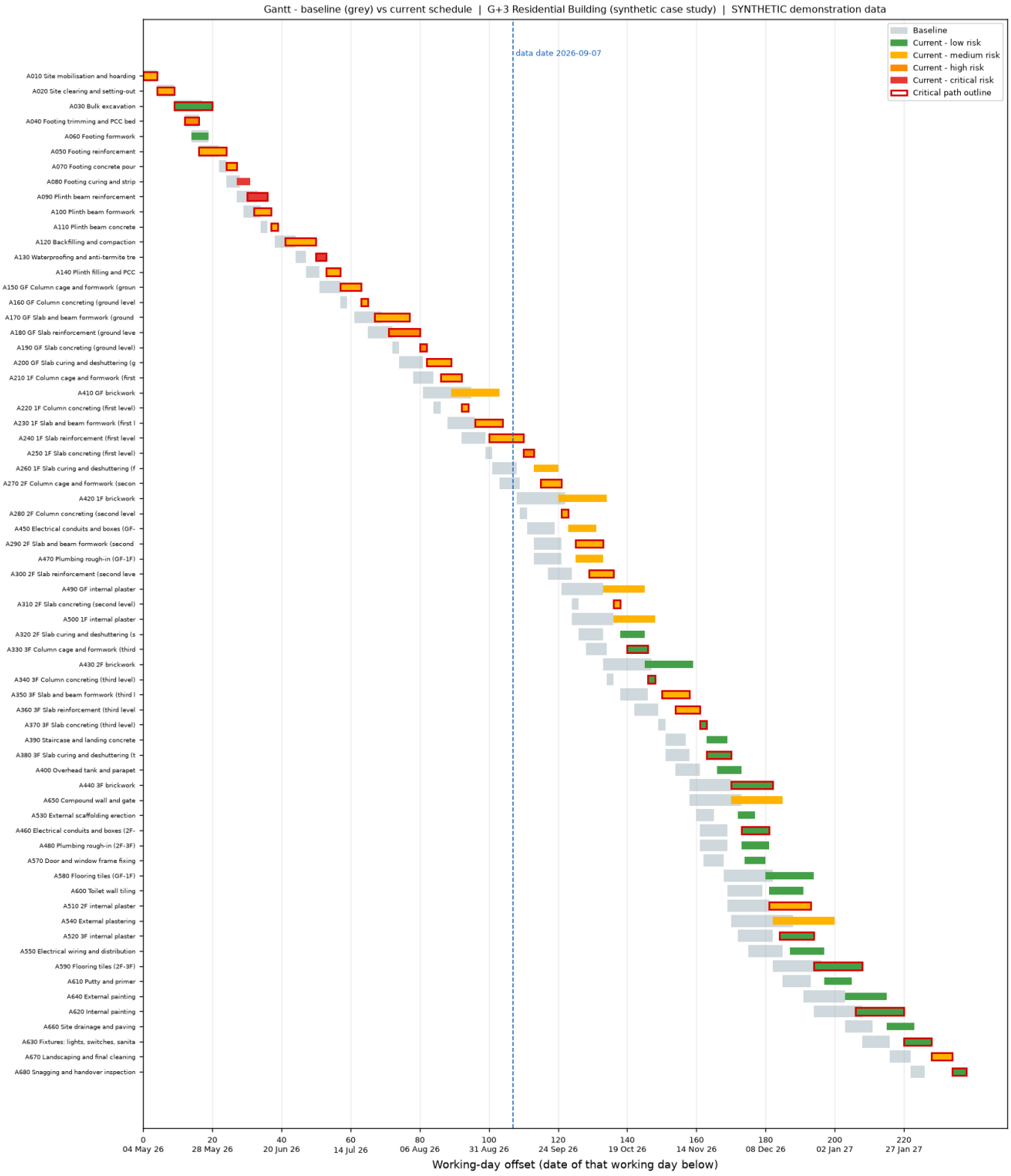
2. Progress status

At data date 2026-09-07 (107 working days elapsed): planned 29.6 %, actual 26.8 %, variance -2.9 percentage points, performance index 0.903 (duration-weighted).



3. Baseline vs current schedule

Baseline completion 2027-02-02; current forecast 2027-02-16 (+12 working days). 8 activities were adapted. Newly critical: none; left critical path: none; near-critical activities: 6; activities with float erosion: 0.



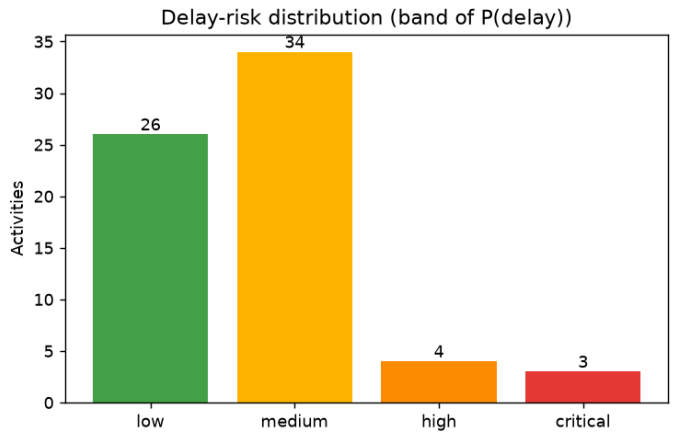
4. Critical path (current schedule)

ID	Activity	Start	Finish	Dur (wd)	Baseline critical
A010	Site mobilisation and hoarding	2026-05-04	2026-05-07	4	yes
A020	Site clearing and setting-out	2026-05-08	2026-05-13	5	yes
A030	Bulk excavation	2026-05-14	2026-05-26	11	yes
A040	Footing trimming and PCC bed	2026-05-18	2026-05-21	4	yes
A050	Footing reinforcement	2026-05-22	2026-06-01	8	yes
A070	Footing concrete pour	2026-06-02	2026-06-04	3	yes
A090	Plinth beam reinforcement	2026-06-09	2026-06-15	6	yes
A100	Plinth beam formwork	2026-06-11	2026-06-16	5	yes
A110	Plinth beam concrete	2026-06-17	2026-06-18	2	yes
A120	Backfilling and compaction	2026-06-22	2026-07-01	9	yes
A130	Waterproofing and anti-termite treatment	2026-07-02	2026-07-04	3	yes
A140	Plinth filling and PCC	2026-07-06	2026-07-09	4	yes
A150	GF Column cage and formwork (ground level)	2026-07-10	2026-07-16	6	yes
A160	GF Column concreting (ground level)	2026-07-17	2026-07-18	2	yes
A170	GF Slab and beam formwork (ground level)	2026-07-22	2026-08-01	10	yes
A180	GF Slab reinforcement (ground level)	2026-07-27	2026-08-05	9	yes
A190	GF Slab concreting (ground level)	2026-08-06	2026-08-07	2	yes
A200	GF Slab curing and deshuttering (ground level)	2026-08-08	2026-08-17	7	yes
A210	1F Column cage and formwork (first level)	2026-08-13	2026-08-20	6	yes
A220	1F Column concreting (first level)	2026-08-21	2026-08-22	2	yes
A230	1F Slab and beam formwork (first level)	2026-08-26	2026-09-03	8	yes
A240	1F Slab reinforcement (first level)	2026-08-31	2026-09-10	10	yes
A250	1F Slab concreting (first level)	2026-09-11	2026-09-15	3	yes
A270	2F Column cage and formwork (second level)	2026-09-18	2026-09-24	6	yes
A280	2F Column concreting (second level)	2026-09-25	2026-09-26	2	yes
A290	2F Slab and beam formwork (second level)	2026-09-30	2026-10-09	8	yes
A300	2F Slab reinforcement (second level)	2026-10-06	2026-10-13	7	yes
A310	2F Slab concreting (second level)	2026-10-14	2026-10-15	2	yes
A330	3F Column cage and formwork (third level)	2026-10-19	2026-10-26	6	yes
A340	3F Column concreting (third level)	2026-10-27	2026-10-28	2	yes
A350	3F Slab and beam formwork (third level)	2026-10-31	2026-11-11	8	yes
A360	3F Slab reinforcement (third level)	2026-11-05	2026-11-14	7	yes
A370	3F Slab concreting (third level)	2026-11-16	2026-11-17	2	yes
A380	3F Slab curing and deshuttering (third level)	2026-11-18	2026-11-25	7	yes
A440	3F brickwork	2026-11-26	2026-12-09	12	yes
A460	Electrical conduits and boxes (2F-3F)	2026-11-30	2026-12-08	8	yes
A510	2F internal plaster	2026-12-09	2026-12-22	12	yes
A520	3F internal plaster	2026-12-12	2026-12-23	10	yes
A590	Flooring tiles (2F-3F)	2026-12-24	2027-01-11	14	yes
A620	Internal painting	2027-01-09	2027-01-25	14	yes
A630	Fixtures: lights, switches, sanitaryware	2027-01-27	2027-02-04	8	yes
A670	Landscaping and final cleaning	2027-02-05	2027-02-11	6	yes
A680	Snagging and handover inspection	2027-02-12	2027-02-16	4	yes

5. At-risk register

ID	Activity	P(delay)	Risk	Pred. delay (d)	Injected (d)	Main drivers
A090	Plinth beam reinforcement	0.80	critical	2.7	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)

ID	Activity	P(delay)	Risk	Pred. delay (d)	Injected (d)	Main drivers
A130	Waterproofing and anti-termite treatment	0.76	critical	1.8	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)
A080	Footing curing and strip	0.74	critical	2.3	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)
A250	1F Slab concreting (first level)	0.67	high	1.3	1	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A180	GF Slab reinforcement (ground level)	0.63	high	3.2	2	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A040	Footing trimming and PCC bed	0.57	high	1.5	0	Financial issues (47%); Quality rework (18%); Equipment breakdown (8%)
A190	GF Slab concreting (ground level)	0.56	high	1.3	0	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A230	1F Slab and beam formwork (first level)	0.47	medium	3.0	0	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A170	GF Slab and beam formwork (ground level)	0.47	medium	3.0	2	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A240	1F Slab reinforcement (first level)	0.46	medium	2.4	3	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A290	2F Slab and beam formwork (second level)	0.46	medium	3.0	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)
A100	Plinth beam formwork	0.45	medium	2.1	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)
A140	Plinth filling and PCC	0.45	medium	1.7	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)
A300	2F Slab reinforcement (second level)	0.44	medium	2.4	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)
A150	GF Column cage and formwork (ground level)	0.44	medium	2.1	0	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A270	2F Column cage and formwork (second level)	0.43	medium	2.1	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)
A410	GF brickwork	0.43	medium	5.3	0	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A210	1F Column cage and formwork (first level)	0.42	medium	2.1	0	Weather delay (42%); Financial issues (30%); Quality rework (11%)
A070	Footing concrete pour	0.41	medium	1.1	1	Financial issues (43%); Weather delay (17%); Quality rework (16%)
A420	1F brickwork	0.41	medium	5.3	0	Weather delay (41%); Financial issues (30%); Quality rework (11%)



6. SHAP explanations

Global importance (mean |SHAP| over this project's activities; explains the model trained on SYNTHETIC data):

Rank	Feature	Mean |SHAP|
1	predecessor_delay_ratio	0.8060
2	f06_financial_issues	0.5396
3	seasonal_exposure	0.4260
4	max_factor_score	0.3386

Rank	Feature	Mean |SHAP|
5	f03_approval_delays	0.3046
6	f09_contractor_performance	0.2851
7	f12_quality_rework	0.2031
8	planned_duration	0.1705
9	type_substructure	0.1510
10	month_cos	0.1245

A090 Plinth beam reinforcement: predecessor_delay_ratio=0.50 (+1.625); seasonal_exposure=1.00 (+0.576); f06_financial_issues=3.00 (+0.540); type_substructure=1.00 (+0.361); max_factor_score=4.00 (+0.339)

A130 Waterproofing and anti-termite treatment: predecessor_delay_ratio=0.50 (+1.625); seasonal_exposure=1.00 (+0.576); f06_financial_issues=3.00 (+0.540); type_substructure=1.00 (+0.361); max_factor_score=4.00 (+0.339)

A080 Footing curing and strip: predecessor_delay_ratio=0.50 (+1.625); seasonal_exposure=1.00 (+0.576); f06_financial_issues=3.00 (+0.540); type_substructure=1.00 (+0.361); max_factor_score=4.00 (+0.339)

A250 1F Slab concreting (first level): predecessor_delay_ratio=0.43 (+1.282); seasonal_exposure=1.00 (+0.576); f06_financial_issues=3.00 (+0.540); max_factor_score=4.00 (+0.339); planned_duration=2.00 (-0.314)

A180 GF Slab reinforcement (ground level): seasonal_exposure=1.00 (+0.576); f06_financial_issues=3.00 (+0.540); predecessor_delay_ratio=0.25 (+0.425); max_factor_score=4.00 (+0.339); f03_approval_delays=3.00 (-0.305)

7. Recommendations

#	Activity	Status	Cause	Recommended action	Recovery (d)	Cost (INR)	Basis
1	A240 1F Slab reinforcement (first level)	critical	All causes (schedule compression)	Crash: compress by 2 working day(s) with extra shifts/overtime	2.0	12,960	crash-limit (cost slope INR 6,480/day, max 2 days)
2	A240 1F Slab reinforcement (first level)	critical	Weather delay	Provide temporary cover/dewatering; resequence weather-sensitive work to dry spells	0.4	3,478	engineering-judgement estimate
3	A240 1F Slab reinforcement (first level)	critical	Financial issues	Secure cash flow: advance or early release of milestone payment	0.4	869	engineering-judgement estimate
4	A240 1F Slab reinforcement (first level)	critical	Quality rework	Introduce pre-pour/pre-plaster QA checklists	0.2	1,739	engineering-judgement estimate
5	A250 1F Slab concreting (first level)	critical	Weather delay	Provide temporary cover/dewatering; resequence weather-sensitive work to dry spells	0.1	690	engineering-judgement estimate
6	A250 1F Slab concreting (first level)	critical	Financial issues	Secure cash flow: advance or early release of milestone payment	0.1	172	engineering-judgement estimate
7	A250 1F Slab concreting (first level)	critical	Quality rework	Introduce pre-pour/pre-plaster QA checklists	0.1	345	engineering-judgement estimate

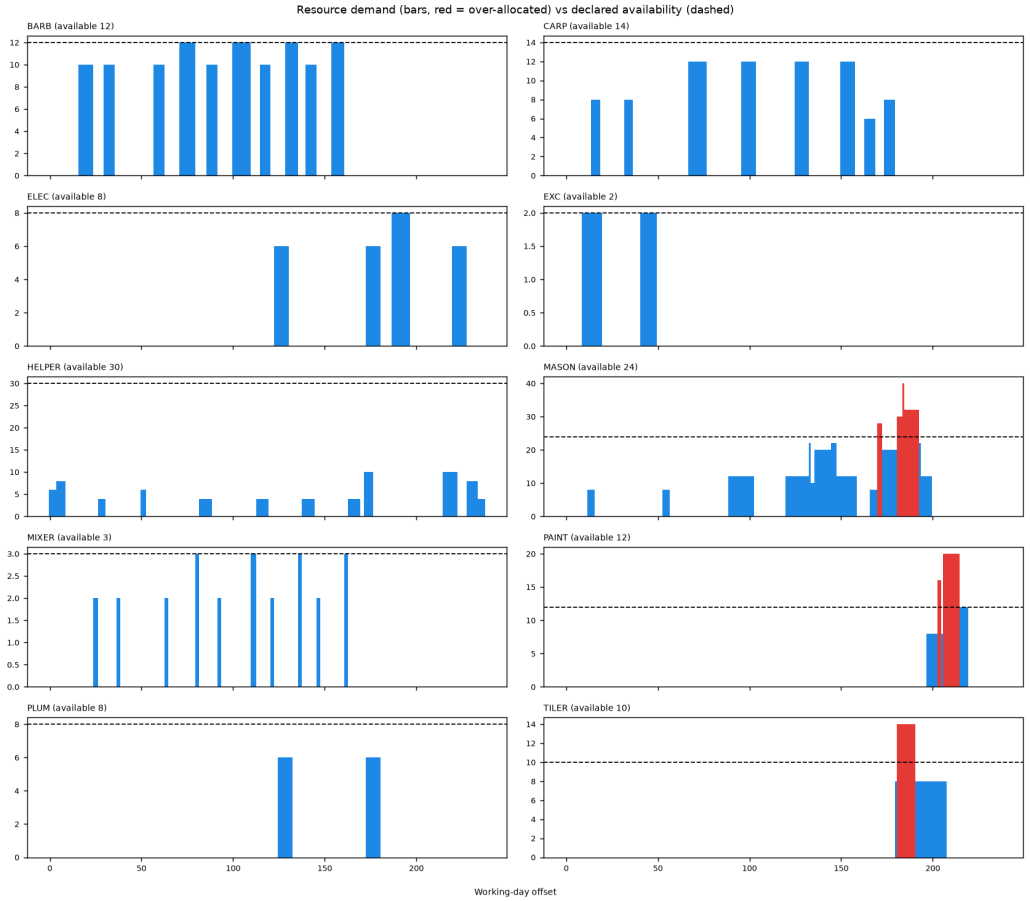
Mitigation estimates are engineering-judgement values; crash estimates follow the crash limits and cost slopes in activities.csv.

8. Resource utilisation

Resource	Available	Peak demand	Mean active util. %	Util. % of project	Over-alloc. days	Shortage (res-days)
BARB	12	12	91.1	27.2	0	0
CARP	14	12	73.0	17.2	0	0
ELEC	8	8	82.4	11.8	0	0
EXC	2	2	100.0	8.4	0	0
HELPER	30	10	20.1	5.7	0	0
MASON	24	40	68.6	27.4	15	110
MIXER	3	3	80.3	7.4	0	0
PAINT	12	20	118.8	11.5	11	80
PLUM	8	6	75.0	5.0	0	0
TILER	10	14	101.4	11.9	10	40

Resource	From	To	Working days	Available	Peak demand	Shortage (res-days)
MASON	2026-11-26	2026-11-28	3	24	28	12
MASON	2026-12-09	2026-12-22	12	24	40	98
PAINT	2027-01-06	2027-01-07	2	12	16	8

Resource	From	To	Working days	Available	Peak demand	Shortage (res-days)
PAINT	2027-01-09	2027-01-19	9	12	20	72
TILER	2026-12-09	2026-12-19	10	10	14	40



9. Methodology and limitations

Pipeline: M1 project/WBS validation -> M2 CPM (NetworkX, FS/SS/FF/SF with lags, integer working-day offsets) -> M3 delay-factor register (15 factors, 0-4 scores, RII-weighted composite risk index) -> M4 delay probability and delay-days models with SHAP -> M5 MAPE-K adaptation (expected value: planned + D x P; conservative: planned + D; only P >= threshold; rounded up; capped) -> M6 ranked recommendations (positive SHAP shares) -> M7 resources and progress -> M8 charts/PDF/web.

The baseline schedule is immutable: every adapted schedule is recomputed from the original durations with duration overrides.

SYNTHETIC DATA: the case study and the model training data are synthetic. Accuracy on synthetic projects is not real-world validation.

RII values are PROVISIONAL (PROVISIONAL - questionnaire survey not yet conducted; not survey-validated).

Several parameters are PROVISIONAL assumptions: risk-band cut-offs, near-critical float of 5 working days, adaptation caps (50 % of planned duration, 30 working days), mitigation effectiveness and cost percentages. They were not taken from a validated source.

Predicted probabilities are Platt-calibrated on synthetic validation projects. Predecessor delay ratio uses observed delays only.

Recommendations are decision support only; cost figures are order-of-magnitude estimates in INR.